



Longitudinal outcomes of body mass index in overweight and obese children with chronic kidney disease

Nancy M. Rodig¹ · Jennifer Roem² · Michael F. Schneider² · Patricia W. Seo-Mayer³ · Kimberly J. Reidy⁴ · Frederick J. Kaskel⁴ · Amy J. Kogon⁵ · Susan L. Furth⁵ · Bradley A. Warady⁶

Received: 1 June 2020 / Revised: 1 December 2020 / Accepted: 19 December 2020
© IPNA 2021

Abstract

Background Longitudinal changes in body mass index (BMI) among overweight and obese children with chronic kidney disease (CKD) are not well characterized. We studied longitudinal trajectories and correlates of these trajectories, as results may identify opportunities to optimize health outcomes.

Methods Longitudinal changes in age-sex-specific BMI z-scores over 1851 person-years of follow-up were assessed in 524 participants of the Chronic Kidney Disease in Children Study. A total of 353 participants were categorized as normal (BMI > 5th to < 85th percentile), 56 overweight (BMI ≥ 85th to 95th percentile) and 115 obese (BMI ≥ 95th percentile) based on the average of three BMI measurements during the first year of follow-up. Studied covariates included age, sex, race, CKD etiology, corticosteroid usage, household income, and maternal education.

Results In unadjusted analysis, BMI z-scores decreased over time in elevated BMI groups (overweight: mean = − 0.06 standard deviations (SD) per year, 95% CI: − 0.11, − 0.01; obese: mean = − 0.04 SD per year, 95% CI: − 0.07, − 0.01). Among obese children, only age was associated with change in BMI z-score; children < 6 years had a mean decrease of 0.19 SD during follow-up (95% CI: − 0.30, − 0.09). Socioeconomic factors were not associated with change in BMI.

Conclusion Overweight and obese children with CKD demonstrated a significant annual decline in BMI, though the absolute change was modest. Among obese children, only age < 6 years was associated with significant decline in BMI. Persistence of elevated BMI in older children and adolescents with CKD underscores the need for early prevention and effective intervention.

Keywords Children · Chronic kidney disease · Overweight · Obesity · Elevated body mass index · Longitudinal

Introduction

Childhood obesity is a well-documented public health crisis and a modifiable risk factor for many adverse health outcomes [1, 2]. Obesity in otherwise healthy children is associated with hypertension, lipid abnormalities, and type 2 diabetes mellitus [3–5]. Furthermore, overweight and obesity during adolescence and young adulthood have been linked with an increased risk for the development of adult onset kidney failure [6, 7]. Similar to the general population, children and adolescents with chronic kidney disease (CKD) demonstrate a high prevalence of overweight and obesity [8]. In the Chronic Kidney Disease in Children (CKiD) study, a large cohort of North American children and adolescents with mild to moderate CKD, one-third of participants were overweight or obese at study entry [9]. Overweight and obese children in the CKiD study were more likely to have multiple concomitant cardiovascular risk factors [10]. Should elevated body mass index

✉ Nancy M. Rodig
nancy.rodig@childrens.harvard.edu

¹ Department of Pediatrics, Boston Children's Hospital, 300 Longwood Ave, Boston, MA 02115, USA

² Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA

³ Department of Pediatrics, Inova Children's Hospital and Pediatric Specialists of Virginia, Falls Church, VA, USA

⁴ Department of Pediatrics, Children's Hospital at Montefiore, Bronx, NY, USA

⁵ Department of Pediatrics, Children's Hospital of Philadelphia, Philadelphia, PA, USA

⁶ Department of Pediatrics, Children's Mercy Kansas City, Kansas City, MO, USA

(BMI) persist during the progression to kidney failure, deleterious effects with respect to access to kidney transplantation and increased mortality have been identified [11].

In the general population, elevated BMI during childhood is a strong predictor of elevated BMI in adulthood [12–14]. Large population-based studies of children in the USA demonstrate the prevalence of overweight and obesity to increase steadily with age [15], and among children aged 5–14 years of age, incident obesity has been found to be highest among the youngest children [16]. Longitudinal changes in BMI among children with CKD and elevated BMI are not well characterized and may not mirror the general population given the effect of disease-related factors including altered appetite [17] and reduced physical activity [18, 19]. In fact, whereas limited studies report longitudinal changes in BMI among children treated with dialysis [20] and kidney transplantation [21, 22], there are no prospective reports evaluating the longitudinal outcomes of BMI in these same children prior to initiating kidney replacement therapy. To address this knowledge gap and identify opportunities for intervention, we studied BMI changes over time and factors associated with these changes among participants in the CKiD study stratified by average BMI level during the first year of study participation (normal, overweight, and obese).

Methods

CKiD is a prospective cohort study of children with mild-to-moderate CKD conducted at over 50 pediatric nephrology centers in North America. The CKiD study design and conduct were approved by an observational study monitoring board appointed by the National Institute of Diabetes and Digestive and Kidney Diseases and by the Institutional Review Boards of each participating center. Details of the study design have been previously published [23]. Briefly, the CKiD study cohort includes more than 1000 participants, 891 of whom were enrolled between April 2005 and March 2014 with a median estimated GFR of 48 mL/min/1.73 m², median age of 11 years, and 62% male. Participants were seen three times during the first year of the study and then annually to provide data on renal, cardiovascular, neurocognitive, and growth characteristics. At each visit in which participants were less than 20 years of age, age-sex-specific BMI z-scores and percentiles were calculated using the 2000 US Centers for Disease Control and Prevention standard growth charts.

Of the 891 CKiD participants, 729 (94%) had a BMI available at three time points in the first year (a baseline visit and two visits occurring on average 6 months apart, the third visit being referred to as the first annual follow-up visit). Of those 729 participants, 589 (81%) had a consistent BMI at all three time points: 396 (67% of 589) had three BMI measures

consistently between the 5th and 85th percentiles and 193 (33% of 589) had three BMI measures consistently greater than the 85th percentile. The BMIs of 140 participants were not consistently within either the normal or elevated range during the first year of study, and these participants were excluded. Finally, 65 of the 589 participants did not have an additional visit after the first annual follow-up visit or complete covariate data at the first annual follow-up visit and were excluded, leaving 524 participants to be included in the analysis. For all analyses, participants were classified into one of three groups based on their average BMI z-score at the three time points over the first year of study participation: 115 obese (BMI $\geq$ 95th percentile), 56 overweight (BMI $\geq$ 85th to $<$ 95th percentile), and 353 normal (BMI $>$ 5th to $<$ 85th percentile).

Demographic and clinical characteristics at the time of the first annual follow-up visit, which occurred at approximately 1 year after study entry, were summarized overall and by BMI group (normal, overweight, obese) using medians and IQRs for continuous variables and percentages for categorical variables. In the first set of analyses, for each BMI group, a linear mixed model was used to quantify the annual change in BMI z-scores over follow-up time after adjusting for the correlation between repeated BMI measurements contributed by the same participant over follow-up. Follow-up time was defined as the first annual follow-up visit through the last study visit in which a participant was less than 20 years of age. In unadjusted models, the slope of the mixed model represents the average annual change in BMI z-score over follow-up time. For each BMI group, a multivariable linear mixed model was used to identify characteristics associated with initial BMI z-score and characteristics associated with change in BMI z-score over follow-up. Each multivariable linear mixed model included first follow-up visit values for age (6–13 or $\geq$ 13 years versus $<$ 6 years): sex, race (Caucasian versus non-Caucasian), household income (\$36,000–\$75,000 or $>$ \$75,000 versus $\leq$ \$36,000), maternal education (at least some college versus high school or less), CKD etiology (glomerular versus non-glomerular), estimated GFR (per 10 mL/min per 1.73 m² decrease) [24], growth hormone use, and corticosteroid use, to assess the main effects of each covariate on initial BMI z-score. In addition to main effect terms for each covariate, we included interaction terms with each covariate and follow-up time in the linear mixed models to allow each covariate to modify the effect on the BMI z-score changes over time.

For those in the elevated (e.g., overweight and obese) BMI groups, we conducted a second analysis, estimating the annual change in each participant's BMI z-score following the first annual follow-up visit by regressing BMI z-score on time from first annual follow-up visit; the slope from each individual's regression model corresponds to each individual's estimated annual change in BMI z-score. We categorized the elevated BMI group into one of three groups depending on

their estimated annual change in BMI z-score, decliners: individuals with estimated annual change < 25th percentile of the distribution of estimated annual changes in BMI z-score; stable: individuals with estimated annual change between the 25th and 75th percentiles; and incliners: individuals with estimated annual change > 75th percentile. A multivariable ordinal logistic regression model was used to quantify the degree of association each demographic and clinical characteristic had with BMI change group (decliners, stable, and incliners). The measure of association between a given characteristic and the outcome (BMI change group) is an odds ratio interpreted as the odds of a worse BMI outcome (higher BMI change category, i.e., incliner versus stable and decliner or incliner and stable versus decliner) among individuals with a given characteristic divided by the odds of a worse BMI outcome among individuals without the characteristic (i.e., with reference values of the characteristic). *P* values < 0.05 were considered statistically significant.

Results

Demographic and clinical characteristics at the time of the first annual follow-up visit, occurring approximately 1 year after study entry, are shown in Table 1 and are presented overall and by BMI group (BMI groups determined by the average BMI z-score during the first year of study as defined in the methods). Overall, only 2% of the study population had edema, with the elevated BMI groups having a low but slightly higher prevalence (4% and 6% for overweight and obese, respectively). Nephrotic range proteinuria was present in 8% of the total population, but serum albumin < 2.5 g/dL was rare among all groups (< 1%). The elevated BMI groups had higher median estimated GFR values. The obese participants were less likely to be Caucasian and had families with lower socioeconomic status. The prevalence of both corticosteroid use and glomerular CKD increased with increasing BMI, and there was also a strong positive relationship between waist circumference z-score and BMI. There were no differences in age, sex, proteinuria, or growth hormone use by BMI group.

Figure 1 depicts each participant's change in BMI z-scores over follow-up by initial BMI categorization group. The 524 participants contributed a total of 1851 person-years of follow-up (median = 3 years; IQR = 2, 5; range = 1, 10). The estimated slope and corresponding 95% confidence interval of the unadjusted linear mixed models are also presented. While BMI z-score decreased over time in each BMI group, the elevated BMI groups showed larger, significant decreases (mean decrease of 0.06 standard deviations (SD) per year, 95% confidence interval (CI): -0.11, -0.01 for overweight and mean decrease of 0.04 SD per year, 95% CI: -0.07, -0.01 for obese). Based upon these data, we estimate that it would take 5.8 years for a child in the overweight group to

attain a BMI < 85th percentile and over 10 years for a child in the obese group to attain a BMI < 95th percentile. Of the 171 participants in either of the elevated BMI groups, only 6% (11) achieved and sustained a BMI below the 85th percentile by the end of study follow-up (for at least the last two visits). Likewise, of the 115 obese children, 10% (12) reached and sustained an overweight BMI (BMI ≥ 85th percentile and < 95th percentile) by the end of study follow-up. Of the 56 participants who started overweight, 11% (6) became and sustained obesity (BMI ≥ 95th percentile) by the end of study follow-up. Of the 353 participants in the normal BMI group, 5% (19) reached and sustained a BMI ≥ 85th percentile by the end of study follow-up.

In a similar set of analyses, but which only included the 180 participants with a baseline eGFR < 45 mL/min/1.73 m², 135 participants who initially had a normal BMI had an average BMI change of -0.05 standard deviations (95% CI: -0.08, -0.01) per year; 15 who were initially overweight had a BMI change of -0.06 standard deviations (95% CI: -0.13, 0.02) per year; and 30 who were initially obese, had a BMI change of -0.07 standard deviations (95% CI: -0.12, -0.02) per year.

Results of the multivariable analysis for each BMI group are presented in Table 2. For each of the BMI groups, the effect of each characteristic on the average BMI z-score during the first year of study participation is shown (top portion of the table), along with the effect of each characteristic on the change in BMI z-score over time (bottom portion of table). The former complements the information presented in Table 1, which describes differences in characteristics between BMI groups, while this analysis shows the effect of each characteristic within each BMI group. Only Caucasian race in the obese group was associated with a lower BMI z-score during the first year of study participation, as Caucasians had an average BMI z-score that was 0.21 SD less than non-Caucasians, 95% CI: -0.36, -0.06. In the group with a BMI between the 5th and 85th percentiles, those who were between 6 and 13 years of age had a significantly lower BMI z-score than those who were < 6 years of age.

BMI z-scores declined significantly over time among those who were initially overweight (BMI decreased on average by 0.15 SD, 95% CI: -0.26, -0.04) or obese (BMI decreased on average by 0.19 SD, 95% CI: -0.30, -0.09). The only characteristics that we examined and that modified these changes were household income in the overweight group and age in the obese group. Among overweight children, household income ≤ \$36,000 was associated with a significant decline in BMI. Among obese children, of the three age groups, only children less than 6 years of age had a significant decline in BMI z-score over time. Though BMI z-scores did not decline significantly during follow-up in the normal BMI group, estimated GFR approximately 1 year after study entry in this group was associated with a change in BMI over time.

Table 1 Characteristics at the first annual follow-up visit for 524 participants, overall and by initial BMI categorization

Characteristic	Overall (<i>n</i> = 524)	Normal BMI percentile (> 5th to < 85th) (<i>n</i> = 353)	Overweight BMI percentile (≥ 85th to < 95th) (<i>n</i> = 56)	Obese BMI percentile (≥ 95th) (<i>n</i> = 115)
Age (years)				
< 6	11%	12%	11%	6%
6–13	43%	44%	36%	44%
≥ 13	46%	44%	53%	50%
Male sex	60%	61%	46%	63%
Race				
Caucasian	69%	73%	68%	56%
African American	20%	17%	20%	31%
Other	11%	10%	12%	13%
Household income				
≤ \$36,000	36%	31%	34%	50%
\$36,000–\$75,000	30%	31%	32%	25%
> \$75,000	34%	37%	34%	25%
Mother had at least some college education	60%	65%	63%	46%
Glomerular diagnosis	29%	23%	30%	46%
Estimated GFR (mL/min/1.73 m ²)	54.6 (39.6, 68.0)	51.3 (37.1, 66.2)	58.2 (43.6, 73.1)	57.9 (43.8, 72.9)
Proteinuria				
< 0.2	38%	39%	27%	42%
0.2–2.0	54%	55%	60%	49%
> 2.0	8%	6%	13%	9%
Albumin < 2.5 g/dL	< 1%	< 1%	0%	0%
Growth hormone use	8%	9%	9%	5%
Corticosteroids use	7%	5%	9%	12%
Edema	2%	1%	4%	6%
Waist Circumference z-score ^a	0.24 (− 0.34, 1.21)	− 0.14 (− 0.55, 0.24)	1.01 (0.68, 1.35)	1.67 (1.34, 2.04)

Median (inter-quartile range) reported for quantitative characteristics, percentages reported for categorical characteristics. Missing data includes proteinuria: *n* = 18; albumin: *n* = 8; edema: *n* = 21; waist circumference z-score: *n* = 280

^a Sharma AK, Metzger DL, Daymont C, Hadjiyannakis S, Rodd CJ (2105) LMS tables for waist-circumference and waist-height ratio Z-scores in children aged 5–16 years in NHANES III: associated with cardio-metabolic risks. *Pediatr Res* 78:723-729

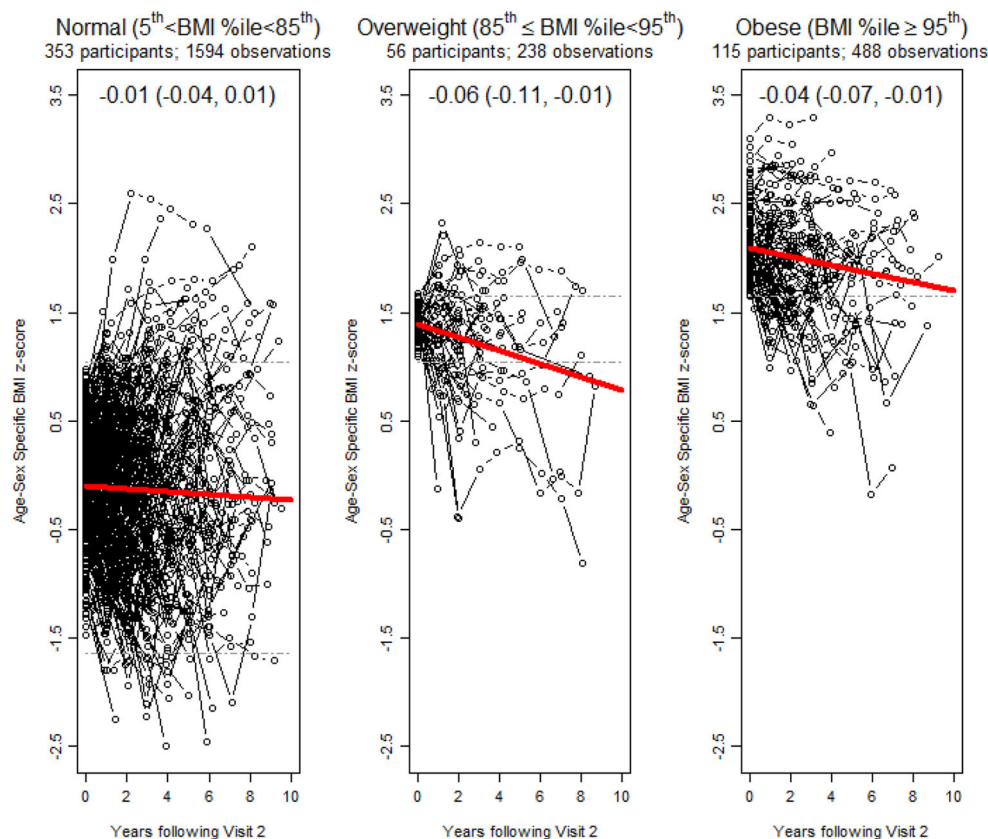
In a second set of analyses, we categorized the group of 171 elevated BMI (i.e., obese and overweight) participants into three groups based on their annual change in BMI z-score: 42 who declined by more than 0.15 SD per year (i.e., < 25th percentile of observed BMI changes) were labeled as decliners, 87 who had a BMI change between − 0.15 SD and 0.05 SD per year (i.e., the 25th and 75th percentiles of observed BMI changes) were labeled as stable, and 42 who increased by more than 0.05 SD per year (i.e., > 75th percentile of observed BMI changes) were labeled as incliners. Results of a multivariable ordinal logistic regression analysis are shown in Table 3. Age and race were the only characteristics that were significantly associated with worse BMI outcomes. Specifically, children who were ≥ 13 years old at the first follow-up visit had 5.09 (95% CI: 1.48, 17.54) higher odds of worse BMI change compared with children who were

< 6 years old at the first follow-up visit. Similarly, Caucasians had 2.05 (95% CI: 1.05, 4.01) higher odds of worse BMI change compared with non-Caucasians. Among the 171 participants with elevated BMI, 45 had a baseline eGFR < 45 mL/min/1.73 m². Compared with the overall group of 171 participants with elevated BMI, there was a slight excess of decliners (33% [*n* = 15 of 45] vs. 25% in overall group) and a slight deficit of both those with a stable BMI (47% [*n* = 21 of 45] vs. 50%) and incliners (20% [*n* = 9 of 45] vs. 25%) among those with elevated BMI and GFR < 45 mL/min/1.73 m².

Discussion

In this study, we observed children with CKD and a normal BMI to have a stable BMI that was only modified by initial

Fig. 1 Individual plots of BMI z-scores over follow-up time by initial BMI categorization. Shown at the top of each panel is the mean and the 95% confidence interval for the average rate of change in BMI z-score per year; the y-intercept (average initial age-sex-specific BMI z-score) and slope (average rate of change in age-sex-specific BMI per year) of each red line were obtained from the corresponding linear mixed model fit to the data



GFR. In contrast, in unadjusted analyses, overweight and obese children demonstrated a significant decline in BMI during longitudinal follow-up, although the absolute change was relatively modest. Furthermore, when we allowed covariates to modify the change in BMI z-scores over time, only age less than 6 years was found to significantly affect BMI changes over time among the obese children. To our knowledge, this is the first study to evaluate the changes in BMI and correlates of these changes over time in overweight and obese children with CKD prior to initiating kidney replacement therapy. These children are unlike otherwise healthy children with an elevated BMI given the potential effects of CKD on health and activity. CKD and declining kidney function have been associated with altered appetite and nutrition, reduced physical activity, and increased sedentary activity [18, 19, 25]. With respect to exposure to medical providers, children with CKD are generally evaluated more frequently than children without CKD, and the recommended evaluation includes assessment of nutritional status [26]. These interactions afford opportunities for counsel regarding the risks associated with elevated BMI and the benefits of healthy diet and exercise with respect to weight reduction [27].

Elevated BMI is associated with negative health outcomes among children with CKD, including increased cardiovascular risks and possibly worse transplant outcomes. In a prior CKiD study, overweight and obese children with CKD

demonstrated higher frequencies of dyslipidemia, abnormal glucose metabolism, and hypertension when compared with children with CKD and a normal BMI [10]. This is particularly important given the impact of cardiovascular disease on the long-term outcome of children with CKD and kidney failure. Furthermore, when associations between BMI at onset of kidney failure and outcomes of transplant and survival were studied, obese children were disadvantaged when compared with children with a normal BMI [11]. Obese children were less likely to receive a kidney transplant and had a lower odds of receiving a living donor kidney when the transplant occurred within 18 months of kidney failure. Additionally, the risk of death over a median follow-up of 7 years was 17% higher among children who were obese when compared with children with a normal BMI at initiation of kidney replacement therapy, although this risk was attenuated when adjusted for the lower transplant rate among obese children. With respect to the association of elevated BMI at the time of transplantation, reported outcomes of allograft and patient survival have varied [28–31].

When lifestyle interventions lead to reduction in BMI among obese children in the general population, health benefits are evident [27, 32]. Even relatively modest decreases in BMI z-score among obese children have been accompanied by improvements in blood pressure, total cholesterol, and aerobic fitness [33]. Given the deleterious health consequences of

Table 2 Multivariable analyses of **effects^a** on baseline BMI z-score at the first annual follow-up visit and change in BMI z-score over follow-up by initial BMI categorization

Characteristic	Normal BMI percentile (> 5th to < 85th) (<i>n</i> = 353; 1594 person-visits)	Overweight BMI percentile (≥ 85th to < 95th) (<i>n</i> = 56; 238 person-visits)	Obese BMI percentile (≥ 95th) (<i>n</i> = 115; 488 person-visits)
<i>Initial BMI Z-score</i>			
Intercept ^b	0.14 (− 0.09, 0.37)	1.40 (0.98, 1.81)	2.32 (2.00, 2.64)
Age (years)			
< 6	0	0	0
6–13	− 0.23 (− 0.43, − 0.03)	− 0.10 (− 0.40, 0.20)	− 0.06 (− 0.34, 0.22)
≥ 13	− 0.19 (− 0.40, 0.01)	− 0.02 (− 0.31, 0.28)	− 0.08 (− 0.37, 0.21)
Male sex	0.01 (− 0.12, 0.14)	0.04 (− 0.16, 0.23)	− 0.01 (− 0.15, 0.13)
Caucasian race	− 0.06 (− 0.21, 0.09)	0.03 (− 0.18, 0.23)	− 0.21 (− 0.36, − 0.06)
Household income			
≤ \$36,000	0	0	0
\$36,000–\$75,000	− 0.06 (− 0.23, 0.11)	0.06 (− 0.18, 0.30)	− 0.03 (− 0.19, 0.14)
> \$75,000	− 0.02 (− 0.19, 0.16)	− 0.08 (− 0.38, 0.23)	− 0.09 (− 0.27, 0.09)
Mother had at least some college education	− 0.02 (− 0.16, 0.12)	0.01 (− 0.27, 0.29)	− 0.09 (− 0.23, 0.06)
Glomerular diagnosis	0.06 (− 0.11, 0.23)	− 0.03 (− 0.32, 0.26)	− 0.01 (− 0.18, 0.15)
GFR centered at 50, per 10 mL/min/1.73 m ² decrease	− 0.01 (− 0.04, 0.03)	0.01 (− 0.05, 0.07)	− 0.03 (− 0.06, 0.003)
Growth hormone use	0.09 (− 0.13, 0.31)	0.08 (− 0.24, 0.39)	− 0.05 (− 0.35, 0.25)
Corticosteroids use	0.21 (− 0.12, 0.53)	− 0.01 (− 0.38, 0.37)	0.07 (− 0.15, 0.29)
<i>Annual change BMI z-score over follow-up</i>			
Years from visit 2 ^c	− 0.01 (− 0.08, 0.06)	− 0.15 (− 0.26, − 0.04)	− 0.19 (− 0.30, − 0.09)
Age (years)			
< 6	0	0	0
6–13	− 0.01 (− 0.07, 0.05)	0.04 (− 0.03, 0.10)	0.15 (0.06, 0.23)
≥ 13	− 0.03 (− 0.10, 0.04)	0.05 (− 0.02, 0.12)	0.15 (0.06, 0.24)
Male sex	− 0.02 (− 0.07, 0.02)	0.02 (− 0.03, 0.07)	− 0.04 (− 0.10, 0.01)
Caucasian race	0.02 (− 0.04, 0.07)	0.04 (− 0.02, 0.11)	0.04 (− 0.01, 0.10)
Household income			
≤ \$36,000	0	0	0
\$36,000–\$75,000	0.00 (− 0.06, 0.05)	0.18 (0.09, 0.27)	− 0.01 (− 0.07, 0.06)
> \$75,000	0.01 (− 0.05, 0.07)	0.13 (0.03, 0.23)	0.01 (− 0.06, 0.08)
Mother had at least some college education	0.03 (− 0.02, 0.07)	− 0.10 (− 0.20, 0.01)	0.03 (− 0.02, 0.08)
Glomerular diagnosis	− 0.06 (− 0.12, 0.01)	− 0.06 (− 0.19, 0.08)	0.00 (− 0.06, 0.06)
GFR centered at 50, per 10 mL/min/1.73 m ² decrease	− 0.02 (− 0.03, − 0.01)	− 0.01 (− 0.03, 0.01)	− 0.01 (− 0.02, 0.01)
Growth hormone use	0.02 (− 0.05, 0.10)	− 0.06 (− 0.17, 0.04)	0.08 (− 0.02, 0.18)
Corticosteroids use	− 0.03 (− 0.18, 0.13)	0.08 (− 0.07, 0.23)	− 0.04 (− 0.13, 0.06)

^a Reported as estimated mean (95% confidence interval)^b Intercept is estimated mean BMI z-score at visit 2 for a child with reference values of all characteristics, i.e., age < 6, female sex, non-Caucasian race, etc...^c Estimated mean change in BMI z-scores over follow-up for a child with reference values of all characteristics, i.e., age < 6, female sex, non-Caucasian race, etc...

Bold indicates the estimate is significant at 0.05 level

elevated BMI among children with CKD, it would be anticipated that a decline in BMI would also be associated with better health outcomes.

With the aim of identifying children at risk who could potentially benefit from intensified counsel and intervention,

we evaluated demographic and disease-related characteristics for associations with elevated BMI and subsequent change in BMI among children with CKD. Among healthy children in the USA, lower socioeconomic status, Hispanic ethnicity, and black race are associated with overweight and obesity [15,

Table 3 Multivariable relationship between BMI categorization over time (decliners [$n = 42$ with annual BMI change < 25 th percentile of estimated BMI changes], stable [$n = 87$ with annual BMI change between 25th and 75th percentile of estimated BMI changes], incliners [$n = 42$ with annual BMI change > 75 th percentile of estimated BMI changes]) and demographic and clinical characteristics among 171 participants having initial overweight or obesity

Characteristic	Odds ratio of a worse categorization of BMI over time (95% confidence interval)
Age (years)	
< 6	1
6–13	3.17 (0.95, 10.56)
≥ 13	5.09 (1.48, 17.54)
Male sex	0.82 (0.44, 1.50)
Caucasian race	2.05 (1.05, 4.01)
Household income	
$\leq \$36,000$	1
\$36,000–\$75,000	1.71 (0.81, 3.64)
$> \$75,000$	1.81 (0.78, 4.17)
Mother had at least some college education	0.76 (0.39, 1.50)
Glomerular diagnosis	1.18 (0.57, 2.45)
GFR centered at 50, per 10 mL/min/1.73 m ² decrease	0.90 (0.77, 1.05)
Growth hormone use	1.67 (0.49, 5.73)
Corticosteroids use	0.56 (0.21, 1.54)

16]. We, therefore, studied the association of BMI and the subsequent change in BMI with traditional risk factors, including race and measures of socioeconomic status. We also studied the effects of disease-related variables in relation to BMI, which are not well reported, including CKD etiology (glomerular versus non-glomerular), estimated GFR, growth hormone use, and corticosteroid use.

Comparable with children in the general population [15, 34], we found that elevated BMI during the first year of study was associated with race and socioeconomic factors in the CKiD cohort. Caucasian race was less common among obese children and was associated with a significantly lower baseline BMI among obese children. Notably, however, Caucasian race was not associated with a subsequent decline in BMI among those who were initially overweight or obese. In fact, when participants with elevated BMI were categorized by their annual change in BMI z-score, Caucasians had higher odds of a worse BMI change over time; this association was not anticipated. To explain this result, one might postulate that Caucasian children with CKD are more sedentary than non-Caucasian children with CKD. It should be noted, however, that a recent large population study found non-Hispanic white children to be less sedentary than Hispanic and non-Hispanic black children [35]. Furthermore, though obese children had

families of lower socioeconomic status, maternal education and household income were not associated with baseline or subsequent change in BMI. Among overweight children, a significant decline in BMI was restricted to those with a household income $\leq \$36,000$. The explanation for this finding is also unclear. Whereas one might hypothesize a contribution of food insecurity, this issue has actually been associated with an increased risk of childhood overweight and obesity in the general population [36, 37]. Whether or not food insecurity has a different impact on children with CKD requires further study.

With respect to disease-related factors, better-preserved kidney function was associated with overweight and obesity. There was also a trend for an association of glomerular disease and corticosteroid use with a higher average BMI during the first year of study. Nearly half of the obese children had a glomerular diagnosis compared with less than 25% of those who had a normal BMI. Glomerular causes of kidney disease are more frequently associated with heavy proteinuria, depressed serum albumin, and fluid retention secondary to reduced oncotic pressure. However, although glomerular disease was present in a substantial percentage of the children with obesity in our cohort, the prevalence of heavy proteinuria was similar across the BMI groups, and hypoalbuminemia (< 2.5 g/dL) was not observed in the elevated BMI groups, suggesting that nephrosis was unlikely to be a significant contributor to the calculated BMI. Volume retention and edema related to other mechanisms, such as reduced glomerular filtration rate, would be expected to artificially increase the calculated BMI. Although overall rates were low (2%), edema was reported more frequently among overweight (4%) and obese (6%) children. The etiology for the higher frequency of edema reported among overweight and obese children is unclear, particularly as kidney function was better preserved in the elevated BMI groups when compared with children with normal BMI. It is plausible that the sodium intake of children with an elevated BMI is higher, accounting for a predisposition to fluid retention. This finding may also reflect, in part, the challenge of making an accurate diagnosis of edema in those with an overweight or obese body habitus. In contrast to glomerular disease, non-glomerular disease includes congenital anomalies of the kidney and urinary tract (CAKUT), which can be associated with polyuria, salt wasting, and relative hypovolemia. Non-glomerular disease was more prevalent among the children with a normal BMI and could have contributed to a decreased measured BMI in some children during the first year of study.

When evaluating disease-related factors and their association with BMI and subsequent change in BMI within each group, only estimated GFR approximately 1 year after study entry in the normal BMI group was associated with a change in BMI over time; each 10 mL/min/1.73 m² lower estimated GFR was associated with a 0.02 SD greater annual decline in BMI during follow-up. Though overweight and obese

children had better preserved kidney function at the beginning of follow-up, kidney function was not associated with subsequent changes in BMI. In particular, lower kidney function was not associated with a significant decline in BMI in those who were initially overweight or obese, despite the recognized negative impact that advanced CKD has been reported to have on appetite and nutritional status. Furthermore, when children with poor kidney function (estimated GFR < 45 mL/min/1.73 m²) were analyzed separately, the annual change of BMI z-score in unadjusted models among all three BMI groups had an average BMI decline that was at least as substantial as the study population as a whole. Children with a normal BMI and poor kidney function demonstrated a significant decline in BMI, though the absolute change was limited (average annual BMI z-score change of −0.05). Underlying diagnosis was also not associated with a change in BMI; children with non-glomerular disease did not have a decline in BMI when compared with those with glomerular kidney disease despite the potential for abnormalities of volume status to influence the BMI determination, as noted above. Of note, the capacity to evaluate the effects of growth hormone and corticosteroid therapy in this analysis was limited as a low percentage (6% and 7%, respectively) of participants were prescribed these treatments.

The relationship of age to the development of overweight and obesity has been examined in the general population with the aim of identifying an optimal window for prevention and intervention. One study observed that half of childhood obesity occurred among children who had become overweight during the preschool years and incident obesity was highest among the youngest children [16]. In our study, age was significantly associated with BMI. Specifically, among obese children, a significant decline in BMI was restricted to children who were < 6 years of age, while older children and adolescents had smaller and non-significant changes in BMI. While there were only seven children who were less than 6 years of age in the obese group, they contributed nearly 10% of the person-visits in this analysis. Moreover, children with an initial elevated BMI who were greater than 13 years of age had a higher odds of a worse BMI change over time when compared with children less than 6 years of age. The reasons for the disparity among younger and older children with CKD in the CKiD cohort are unclear. Though the younger children would be more likely to have non-glomerular disease, glomerular disease was not associated with a change in BMI. One could hypothesize that an important factor may be the frequent use of enteral feeds in very young children with CKD, resulting in overfeeding and an elevated BMI [38], which “normalizes” when enteral feeds are discontinued. As for the older children, they appear to be more likely to experience persistent obesity and the associated metabolic and cardiovascular risks [10, 39], likely secondary to a preference for sedentary activities, such as video games and computer use, in contrast to physical exercise [19], as well as less than optimal

nutrition. Our findings of persistently elevated BMI in older children emphasize the importance of close nutritional management throughout the course of CKD with the goal of maintaining a healthy weight and preventing the development of overweight and obesity. Given that cardiovascular disease is a leading cause of morbidity and mortality among children and adults with kidney failure [40, 41], addressing modifiable risk factors, such as elevated BMI, is of paramount importance.

This analysis of BMI in the CKiD cohort has numerous strengths, including cohort size, availability of multiple assessments for each subject, and accurate measurements of growth parameters. However, limitations must be acknowledged. Data regarding nutritional intake and food insecurity is lacking, and we were unable to study the effects of interventions such as nutritional plans, exercise programs, and referrals to subspecialty clinics addressing obesity. Additionally, the population enrolled may differ from the general population of children with CKD and thus subject the data to selection bias.

In conclusion, overweight and obese children in this cohort of individuals with mild to moderate CKD demonstrated a significant decline in BMI during follow-up, although the absolute change was relatively modest. Among obese children, a decline in BMI was only observed in children less than 6 years of age. Moreover, among children with an initial elevated BMI, age greater than 13 years was associated with higher odds of worse BMI change when compared with children less than 6 years of age. The relative persistence of obesity in older children and adolescents with CKD is sobering and underscores the importance of prevention and the need to address elevated BMI in early childhood. Given the frequency that children with CKD are evaluated by care providers, an elevated BMI or a concerning trend of BMI change should prompt counsel, ideally with a dietitian during these visits regarding the associated long-term health risks and the recommendation for lifestyle changes to optimize health outcomes.

Acknowledgments Data in this manuscript were collected by the Chronic Kidney Disease in Children (CKiD) study with clinical coordinating centers (Principal Investigators) at Children’s Mercy Hospital and the University of Missouri, Kansas (B.W.) and Children’s Hospital of Philadelphia (Susan Furth); central biochemistry laboratory at the University of Rochester Medical Center (George Schwartz); and data coordinating center at the Johns Hopkins Bloomberg School of Public Health (Derek Ng, Alvaro Munõz).

Authors’ contributions Research idea and study design: NR, JR, MS, BW; data acquisition: JR, MS; data analysis/interpretation: NR, JR, MS, PS, KR, FK, AK, SF, BW; statistical analysis: JR, MS; supervision or mentorship: BW. Each author contributed important intellectual content during manuscript drafting or revision and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved.

Funding The CKiD Study is supported by grants from the National Institute of Diabetes and Digestive and Kidney Diseases, with additional

funding from the Eunice Kennedy Shriver National Institute of Child Health and Human Development, and the National Heart, Lung, and Blood Institute (U01-DK-66143, U01-DK-66174, U24-DK-082194, U24-DK-66116). The CKiD website is located at <https://statepi.jhsph.edu/ckid>.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

- NCD Risk Factor Collaboration (2017) Worldwide trends in body-mass index, underweight, overweight, and obesity from 1975 to 2016: a pooled analysis of 2416 population-based measurement studies in 128.9 million children, adolescents, and adults. *Lancet* 390:2627–2642. [https://doi.org/10.1016/S0140-6736\(17\)32129-3](https://doi.org/10.1016/S0140-6736(17)32129-3)
- Ogden CL, Carroll MD, Lawman HG, Fryar CD, Kruszon-Moran D, Kit BK, Flegal KM (2016) Trends in obesity prevalence among children and adolescents in the United States, 1988–1994 through 2013–2014. *JAMA* 315:2292–2299
- Maggio ABR, Aggoun Y, Marchand LM, Martin XE, Hermann F, Beghetti M, Farpour-Lambert NJ (2008) Associations among obesity, blood pressure, and left ventricular mass. *J Pediatr* 152:489–493. <https://doi.org/10.1016/j.jpeds.2007.10.042>
- Williams DP, Going SB, Lohman TG, Harsha DW, Srinivasan SR, Webber LS, Berenson GS (1992) Body fatness and risk for elevated blood pressure, total cholesterol, and serum lipoprotein ratios in children and adolescents. *Am J Public Health* 152:358–363
- Pinhas-Hamiel O, Dolan LM, Daniels SR, Standiford D, Khoury PR, Zeitler P (1996) Increased incidence of non-insulin-dependent diabetes mellitus among adolescents. *J Pediatr* 128:608–615. [https://doi.org/10.1016/S0022-3476\(96\)80124-7](https://doi.org/10.1016/S0022-3476(96)80124-7)
- Vivante A, Golan E, Tzur D, Leiba A, Tirosh A, Skorecki K, Calderon-Margalit R (2012) Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med* 172:1644–1650. <https://doi.org/10.1001/2013.jamainternmed.85>
- Hsu CY, McCulloch CE, Iribarren C, Darbinian J, Go AS (2006) Body mass index and risk for end-stage renal disease. *Ann Intern Med* 144:21–28. <https://doi.org/10.7326/0003-4819-144-1-200601030-00006>
- Filler G, Reimão SM, Kathiravelu A, Grimmer J, Feber J, Drukker A (2007) Pediatric nephrology patients are overweight: 20 years' experience in a single Canadian tertiary pediatric nephrology clinic. *Int Urol Nephrol* 39:1235–1240. <https://doi.org/10.1007/s11255-007-9258-y>
- Rodig NM, McDermott KC, Schneider MF, Hotchkiss HM, Yadin O, Seikaly MG, Furth SL, Warady BA (2014) Growth in children with chronic kidney disease: a report from the Chronic Kidney Disease in Children Study. *Pediatr Nephrol* 29:1987–1995. <https://doi.org/10.1007/s00467-014-2812-9>
- Wilson AC, Schneider MF, Cox C, Greenbaum LA, Saland J, White CT, Furth S, Warady BA, Mitsnefes MM (2011) Prevalence and correlates of multiple cardiovascular risk factors in children with chronic kidney disease. *Clin J Am Soc Nephrol* 6:2759–2765. <https://doi.org/10.2215/CJN.03010311>
- Ku E, Glidden DV, Hsu CY, Portale AA, Grimes B, Johansen KL (2016) Association of body mass index with patient-centered outcomes in children with ESRD. *J Am Soc Nephrol* 27:551–558. <https://doi.org/10.1681/ASN.2015010008>
- Wang LY, Chyen D, Lee S, Lowry R (2008) The association between body mass index in adolescence and obesity in adulthood. *J Adolesc Heal* 42:512–518. <https://doi.org/10.1016/j.jadohealth.2007.10.010>
- Deshmukh-Taskar P, Nicklas TA, Morales M, Yang SJ, Zakeri I, Berenson GS (2006) Tracking of overweight status from childhood to young adulthood: the Bogalusa Heart Study. *Eur J Clin Nutr* 60:48–57. <https://doi.org/10.1038/sj.ejcn.1602266>
- Juhola J, Magnussen CG, Viikari JSA, Kahonen M, Hutri-Kahonen N, Jula A, Lehtimäki T, Akerblom HK, Pietikainen M, Laitinen T, Jokinen E, Taittonen L, Raitakari OT, Juonala M (2011) Tracking of serum lipid levels, blood pressure, and body mass index from childhood to adulthood: the Cardiovascular Risk in Young Finns Study. *J Pediatr* 159:584–590. <https://doi.org/10.1016/j.jpeds.2011.03.021>
- Skinner AC, Ravanbakht SN, Skelton JA, Perrin EM, Armstrong SC (2018) Prevalence of obesity and severe obesity in US children, 1999–2016. *Pediatrics* 141:e20173459
- Cunningham SA, Kramer MR, Narayan KMV (2014) Incidence of childhood obesity in the United States. *N Engl J Med* 370:403–411. <https://doi.org/10.1056/nejmoa1309753>
- Ayestaran FW, Schneider MF, Kaskel FJ, Srivaths PR, Seo-Mayer PW, Moxey-Mims M, Furth SL, Warady BA, Greenbaum LA (2016) Perceived appetite and clinical outcomes in children with chronic kidney disease. *Pediatr Nephrol* 31:1121–1127. <https://doi.org/10.1007/s00467-016-3321-9>
- Akber A, Portale AA, Johansen KL (2012) Pedometer-assessed physical activity in children and young adults with CKD. *Clin J Am Soc Nephrol* 7:720–726. <https://doi.org/10.2215/CJN.06330611>
- Clark SL, Denburg MR, Furth SL (2016) Physical activity and screen time in adolescents in the chronic kidney disease in children (CKiD) cohort. *Pediatr Nephrol* 31:801–808. <https://doi.org/10.1007/s00467-015-3287-z>
- Schaefer F, Benner L, Borzych-Dużalka D, Zaritsky J, Xu H, Rees L, Antonio ZL, Serdaroglu E, Hooman N, Patel H, Sever L, Vondrak K, Flynn J, Rebori A, Wong W, Holtta T, Yildirim ZY, Ranchin B, Grenda R, Testa S, Drozd D, Szabo AJ, Eid L, Basu B, Vitkevicius R, Wong C, Pottoré SJ, Müller D, Dunsen R, Cledon CG, Fila M, Sartz L, Sander A, Warady BA, International Pediatric Peritoneal Dialysis Network (2019) Global variation of nutritional status in children undergoing chronic peritoneal dialysis: a longitudinal study of the International Pediatric Peritoneal Dialysis Network Registry. *Sci Rep* 9:4886. <https://doi.org/10.1038/s41598-018-36975-z>
- Sgambati K, Clauss S, Lei KY, Song J, Rahaman SO, Lasota M, Moudgil A (2018) Effects of obesity and metabolic syndrome on cardiovascular outcomes in pediatric kidney transplant recipients: a longitudinal study. *Pediatr Nephrol* 33:1419–1428. <https://doi.org/10.1007/s00467-017-3860-8>
- Plumb LA, Pitcher D, Tse Y, Shield JP, Inward C, Sinha MD, British Association for Paediatric Nephrology (2014) Longitudinal changes in body mass index following renal transplantation in UK children. *Nephrol Dial Transplant* 29:196–203. <https://doi.org/10.1093/ndt/gfz395>
- Furth SL, Abraham AG, Jerry-Fluker J, Schwartz GJ, Benfield M, Kaskel F, Wong C, Mak RH, Moxey-Mims M, Warady BA (2011) Metabolic abnormalities, cardiovascular disease risk factors, and GFR decline in children with chronic kidney disease. *Clin J Am Soc Nephrol* 6:2132–2140. <https://doi.org/10.2215/CJN.07100810>
- Schwartz GJ, Schneider MF, Maier PS, Moxey-Mims M, Dharnidharka VR, Warady BA, Furth SL, Munoz A (2012) Improved equations estimating GFR in children with chronic kidney disease using an immunonephelometric determination of cystatin C. *Kidney Int* 82:445–453. <https://doi.org/10.1038/ki.2012.169>
- Chen W, Ducharme-Smith K, Davis L, Hui WF, Warady BA, Furth SL, Abraham AG, Betoko A (2017) Dietary sources of energy and

- nutrient intake among children and adolescents with chronic kidney disease. *Pediatr Nephrol* 32:1233–1241. <https://doi.org/10.1007/s00467-017-3580-0>
26. KDOQI Work Group (2009) KDOQI clinical practice guideline for nutrition in children with CKD: 2008 update. Executive Summary. *Am J Kidney Dis* 53:S11–S104
 27. Ho M, Garnett SP, Baur LA, Burrows T, Stewart L, Neve M, Collins C (2013) Impact of dietary and exercise interventions on weight change and metabolic outcomes in obese children and adolescents a systematic review and meta-analysis of randomized trials. *JAMA Pediatr* 167:759–768
 28. Amaral S, Sayed BA, Kutner N, Patzer RE (2016) Preemptive kidney transplantation is associated with survival benefits among pediatric patients with end-stage renal disease. *Kidney Int* 90:1100–1108. <https://doi.org/10.1016/j.kint.2016.07.028>
 29. Kaur K, Jun D, Grodstein E, Singer P, Castellanos L, Teperman L, Molmenti E, Fahmy A, Frank R, Infante L, Sethna CB (2018) Outcomes of underweight, overweight, and obese pediatric kidney transplant recipients. *Pediatr Nephrol* 33:2353–2362. <https://doi.org/10.1007/s00467-018-4038-8>
 30. Dick AAS, Hansen RN, Montenegro MI, Healey PJ, Smith JM (2017) Body mass index as a predictor of outcomes among pediatric kidney transplant recipient. *Pediatr Transplant* 21. <https://doi.org/10.1111/ptr.12992>
 31. Hanevold CD, Ho PL, Talley L, Mitsnefes MM (2005) Obesity and renal transplant outcome: a report of the North American Pediatric Renal Transplant Cooperative Study. *Pediatrics* 115:352–356. <https://doi.org/10.1542/peds.2004-0289>
 32. Ho M, Garnett SP, Baur L, Burrows T, Stewart L, Neve M, Collins C (2012) Effectiveness of lifestyle interventions in child obesity: systematic review with meta-analysis. *Pediatrics* 130:e1647–e1671
 33. Kirk S, Zeller M, Claytor R, Santangelo M, Khoury PR, Daniels SR (2005) The relationship of health outcomes to improvement in BMI in children and adolescents. *Obes Res* 13:876–882. <https://doi.org/10.1038/oby.2005.101>
 34. Skinner AC, Skelton JA (2014) Prevalence and trends in obesity and severe obesity among children in the United States, 1999–2012. *JAMA Pediatr* 168:561–566. <https://doi.org/10.1001/jamapediatrics.2014.21>
 35. Yang L, Cao C, Kantor ED, Nguyen LH, Zheng X, Park Y, Giovannucci EL, Matthews CE, Colditz GA, Cao Y (2019) Trends in sedentary behavior among the US Population, 2001–2016. *JAMA* 321:1587–1597. <https://doi.org/10.1001/jama.2019.3636>
 36. Casey PH, Simpson PM, Gossett JM, Bogle ML, Champagne CM, Connell C, Harsha D, McCabe-Sellers B, Robbins JM, Stuff JE, Weber J (2006) The association of child and household food insecurity with childhood overweight status. *Pediatrics* 118:e1406–e1413. <https://doi.org/10.1542/peds.2006-0097>
 37. Lee AM, Scharf RJ, DeBoer MD (2018) Association between kindergarten and first-grade food insecurity and weight status in U.S. children. *Nutrition* 51–52:1–5. <https://doi.org/10.1016/j.nut.2017.12.008>
 38. Rees L, Azocar M, Borzych D, Watson AR, Buscher A, Edefonti A, Bilge I, Askenazi D, Leozappa G, Gonzales C, van Hoeck K, Secker D, Zurowska A, Ronnholm K, Bouts AH, Stewart H, Ariceta G, Ranchin B, Warady BA, Schaefer F, International Pediatric Peritoneal Dialysis Network (IPPN) Registry (2011) Growth in very young children undergoing chronic peritoneal dialysis. *J Am Soc Nephrol* 22:2303–2312. <https://doi.org/10.1681/ASN.2010020192>
 39. Schaefer F, Doyon A, Azukaitis K, Bayazit A, Canpolat N, Duzova A, Niemirska A, Sozeri B, Thurn D, Anarat A, Ranchin B, Litwin M, Caliskan S, Candan C, Baskin E, Yilmaz E, Mir S, Kirchner M, Sander A, Haffner D, Melk A, Wuhl E, Shroff R, Querfeld U, 4C Study Consortium (2017) Cardiovascular phenotypes in children with CKD: the 4C study. *Clin J Am Soc Nephrol* 12:19–28. <https://doi.org/10.2215/CJN.01090216>
 40. Groothoff JW (2005) Long-term outcomes of children with end-stage renal disease. *Pediatr Nephrol* 20:849–853
 41. Modi ZJ, Lu Y, Ji N, Kapke A, Selewski DT, Dietrich X, Abbott K, Nallamothu BK, Schaubel DE, Saran R, Gipson DS (2019) Risk of cardiovascular disease and mortality in young adults with end-stage renal disease: an analysis of the US Renal Data System. *JAMA Cardiol* 4:353–362. <https://doi.org/10.1001/jamacardio.2019.0375>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.